SOFTWARE ENGINEERING

LAB 05

ISSUE TABLE

Issue	Type	Line(s)	Description	Fix Approach
Hardcoded password string	Security (Bandit - B105)	5	BANK_SYNC_PASSWORD assigned fixed string directly	Retained fallback but dynamically read from os.getenv()
Hardcoded temp directory	Security (Bandit - B108)	8	/tmp/expense_reports hardcoded path	Replaced with cross-platform tempfile.gettempdir()
Deprecated mktemp()	Security (Bandit - B306)	55	tempfile.mktemp() is unsafe against race conditions	Replaced with tempfile.NamedTemporaryFile() context
Command execution via shell	Security (Bandit - B605)	61	os.system("xdg-open " + path) vulnerable to shell injection	Replaced with list-based subprocess.run(["xdg-open", path])
Invalid Naming Convention	Style (Pylint - C0103)	11, 20, 28, 33, 38, 44, 49, 53, 59	Functions and variables used camelCase naming	Converted all function/variable names to Python standard snake_case
Missing Docstrings	Code Quality (Pylint - C0114, C0116)	1, 11, 20, 28, 33, 38, 44, 49, 53, 59, 64	Missing module and function docstrings	Added clear single-line docstrings to module and all functions
Broad Exception Catch	Bug/Quality (Pylint - W0718)	41	except Exception: caught all execution exceptions blindly	Removed broad try-except block and replaced with explicit guard check if not expenses:

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PEP 8 Formatting	Style (Flake8 - E203, E302, E261, E501)	15, 44, 74, 77	Spacing before colons, inline comment spacing, line length over 79 characters	Fixed whitespace around colons, ensured two blank lines, and split long print call across multiple lines

REPO LINK:

https://github.com/Sumeet-ctrl21/L5_32_expense_tracker